

**HALDIA INSTITUTE OF TECHNOLOGY**

(AN AUTONOMOUS INSTITUTION UNDER MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL)

Paper Code : PCC-CS 302

Paper Name : Formal Language & Automata Theory

Time Allotted : 3 Hours

Full Marks : 70

*The figures in the margin indicate full marks**Candidates are required to give their answers in their own words as far as practicable***Group-A****(Multiple Choice Type Questions)**

1. Choose the correct alternatives from the followings:

[15 x 1 = 15]

(I) Which of the following is not a part of 5-tuple finite automata?

CO-2

- (a) Input alphabet
- (b) Transition function
- (c) Initial State
- (d) Output Alphabet

(II) PDA is more powerful than

CO-3

- (a) Turing machine
- (b) Multi tape Turing machine
- (c) Finite automata
- (d) All of these

(III) If we select a string w such that $w \in L$, and $w=xyz$. Which of the following portions cannot be an empty string?

CO-2

- (a) x
- (b) y
- (c) z
- (d) All of the mentioned

(IV) Turing Machine can update symbols on its tape, whereas the Finite Automata cannot update symbols on tape.

CO-4

- (a) correct
- (b) incorrect
- (c) can't say
- (d) may be

(V) If we have more than one accepting states or an accepting state with an out degree, which of the following actions will be taken?

CO-3

- (a) Addition of new state
- (b) Removal of a state
- (c) Make the newly added state as final
- (d) More than one option is correct

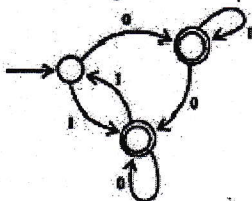
(VI) Recursive language is also called

CO-4

- (a) turing recognizable language
- (b) turing decidable language
- (c) turing recognizable and decidable language
- (d) none of these

(VII) Which string is not accepted by the following FSA (Finite state automata) ?

CO-2



(a) 00111

- (c) 01010
(d) 11010
- (VIII) Recursive language is not closed under
- (a) Union
(b) Intersection
(c) Complement
(d) Intersection
- (IX) Which of the following production is in CNF?
- (a) $S \rightarrow aA$
(b) $SA \rightarrow AS$
(c) $S \rightarrow AB$
(d) All of these
- (X) A regular grammar can be converted to which of the following automata?
- (a) Deterministic Turing Machine
(b) Pushdown Automaton
(c) Finite Automaton
(d) None of the above
- (XI) Which of the following can describe regular languages?
- (a) Context-free grammar
(b) Finite automata
(c) Turing machines
(d) Stack automata
- (XII) The pumping lemma is often used to prove that a language is:
- (a) Context free
(b) Not context free
(c) Regular
(d) None of the mentioned
- (XIII) Extended transition function is
- (a) $Q * \Sigma^* \rightarrow Q$
(b) $Q * \Sigma \rightarrow Q$
(c) $Q^* * \Sigma^* \rightarrow \Sigma$
(d) $Q * \Sigma \rightarrow \Sigma$
- (XIV) The configuration or instantaneous description of PDA is has the following elements
- (a) Current state
(b) Stack content
(c) Unconsumed input
(d) All of the mentioned
- (XV) If L is a regular language, which of the following is guaranteed to be regular?
- (a) Complement of L
(b) Intersection of L with another regular language
(c) Concatenation of L with another regular language
(d) All of the above

CO-4

CO-3

CO-1

CO-1

CO-2

CO-1

CO-3

CO-1

Group-B

(Short Answer Type Questions)

[3 x 5 = 15]

- 2(I) Describe the process of converting a regular expression to a finite automaton (DFA or NFA). Provide an example to illustrate the conversion.

[5] CO-1

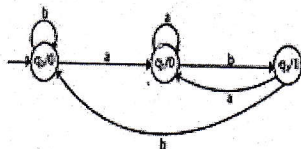
OR

- 2(II) (a) What is regular expression? Write the regular expression for all strings over $\Sigma = \{a, b, c\}$ with exactly one a.

[3] CO-1

3(I) Convert the following Moore machine into Mealy machine

[5] CO-2



OR

3(II) Construct a DFA which accepts set of all string containing even number of 0's and even number of 1's. Also draw its transition table.

[5] CO-2

4(I) Remove useless production from the following CFG

[5] CO-3

$S \rightarrow aA \mid B \mid C,$
 $A \rightarrow aA \mid \lambda,$
 $B \rightarrow Aa,$
 $C \rightarrow aCD,$
 $D \rightarrow addd$

OR

4(II) Write Context Free Grammar for binary palindromes (odd and even both)

[5] CO-3

Group-C

(Long Answer Type Questions)

[4 x 10 = 40]

5(I) Construct DFA for the language $L(G) = \{a^n b : n \geq 0\}$

[10] CO-1

OR

5(II) Demonstrate the equivalence between regular expressions and finite automata by converting the regular expression $a^*b^* + ba^*$ into an NFA and then to a DFA.

[10] CO-1

6(I) (a) Construct finite automata (FA) equivalent to the regular expression

[5] CO-2

$ab(a+b)(ab)^*b$

Show step-by-step methods.

Construct finite automata (FA) equivalent to the regular expression

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Show step-by-step methods

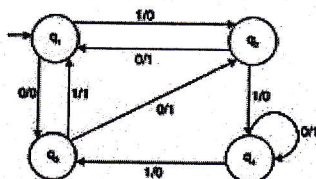
(b) Construct DFA for the language $L(G) = \{a^n b : n \geq 0\}$. Write down five differences between Moore machine and Mealy machine.

[5]

OR

6(II) (a) Convert the given Mealy machine to an equivalent Moore machine

[7] CO-2



(b) what are the differences between mealy machine and moore machine?

[3]

7(I) (a) Convert the following Mealy machine to equivalent Moore machine

[8] CO-3

Present State	Input=0		Input=1	
	Next state	Output	Next state	Output
$\rightarrow q_0$	q_1	0	q_2	0
q_1	q_1	0	q_2	1
q_2	q_1	1	q_2	0

(b) What do you mean by moore machine. explain with suitable example.

[2]

OR

7(II) (a) Write short note on DPDA.

[5] CO-3

(b) Write short note on NPDA.

[5]

8(I) (a) Write short note on Turing recognizable language.

[5] CO-4

(b) Write short note on Turing decidable language.

[5]

OR